Review

Measure B $\Rightarrow$

$\hat{B}\psi = b_k\psi \Rightarrow \boldsymbol{b_k}$

$\hat{B}\psi \neq b_k\psi \Rightarrow$ **Get one of b_i s**

Postulate 3. The only possible values that can result from measurements of the physically observed property B are **the eigenvalues b_i** in the equation $\hat{B}g_i = b_i g_i$, where $\hat{B}$ is the operator corresponding to the property B. **The eigenfunctions g_i are required to be well-behaved.**

If Ψ is an eigenfunction of $\hat{B}$ with eigenvalue b_k, then a measurement of B is certain to yield the value b_k.

Review

Measure B $\Rightarrow$

$\hat{B}\psi = b_k\psi \Rightarrow b_k$

$\hat{B}\psi \neq b_k\psi \Rightarrow$ **Get one of b_i s,**

but which one? $\Rightarrow$ Probability

Postulate 4. If $\hat{B}$ is any linear Hermitian operator that represents a physically observable property, then the eigenfunctions g_i of $\hat{B}$ form a complete set.

Postulate 5. If $\Psi(q,t)$ is the normalized state function of a system at time t, then the **average value** of a physical observable B at time t is

$$\langle B \rangle = \int \Psi^* \hat{B} \Psi d\tau$$

The superposition of states

$$\Psi = \sum_i c_i g_i$$

$$\langle B \rangle = \sum_i |c_i|^2 b_i$$

Review

The Variation Theorem.

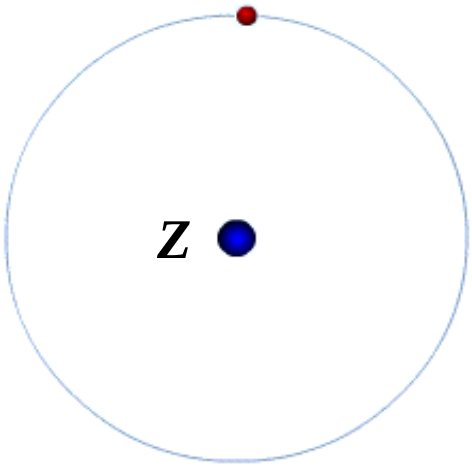
Given a system whose Hamiltonian operator $\hat{H}$ is time independent and whose lowest-energy eigenvalue is E_1, if **ϕ is any normalized well-behaved function** of the coordinates of the system's particles that satisfies the boundary conditions of the problem, then

$$\int \phi^* \hat{H} \phi d\tau \geq E_1$$

If **ϕ is not normalized,** **the variation theorem:**

$$\frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} \geq E_1$$

Hydrogenic (hydrogenlike) Atoms 类氢原子



A hydrogenic atom:
a one-electron atom or one-electron ion of general atomic number *Z*;
e.g., H, He^{+}, Li^{2+},O^{7+}, U^{91+}…

- Their Schrödinger equations can be solved exactly;
- Provide a set of concepts to describe the structures of many-electron atoms and molecules.

The Hamiltonian for a hydrogenic atom:

$$\hat{H} = \hat{E}_{\mathrm{K,electron}} + \hat{E}_{\mathrm{K,nucleus}} + \hat{V}$$

$$= -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_N}\nabla_N^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$$

Convert to center-of-mass coordinates

Separate to two equations

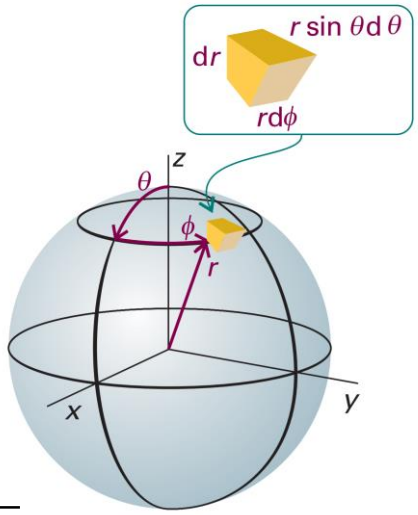
- **The motion of the atom as a whole**
- **The motion of the electron relative to the nucleus**

1-*e* Schrödinger equation in spherical polar coordinates:

$$\left(\frac{-\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)\ ,$$

$$\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_N}$$

The reduced mass



$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

Hydrogenic Atoms

1-*e* Schrödinger equation in spherical polar coordinates:

$$\left(\frac{-\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underbrace{\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}}_{\frac{1}{r^2}\Lambda^2}$$

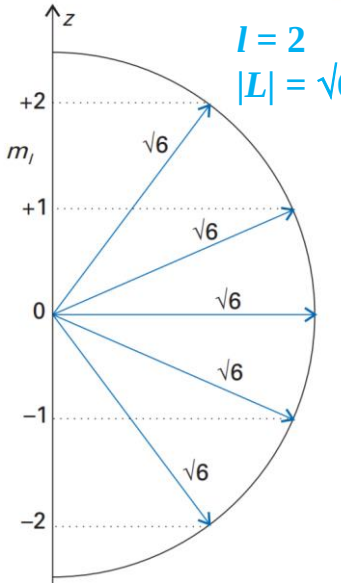
$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

$$\rightarrow \begin{cases} \Lambda^2 Y = -l(l+1)Y & \text{Angular part} \\ -\dfrac{\hbar^2}{2\mu}\dfrac{\mathrm{d}^2u}{\mathrm{d}r^2} + V_{\text{eff}}u = Eu \qquad u = rR & \text{Radial part} \end{cases}$$

$$V_{\text{eff}} = -\frac{Ze^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2\mu r^2}$$

Angular wave equation

$$\Lambda^2 Y = -l(l+1)Y$$



$l = 2$
$|L| = \sqrt{6}\hbar$

l, m specify the angular momentum of the electron around the nucleus.

l,	m,	$Y_{l,m}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$
1	0	$\left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$
	±1	$\mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\, e^{\pm i\phi}$
2	0	$\left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta-1)$
	±1	$\mp\left(\frac{15}{8\pi}\right)^{1/2}\cos\theta\sin\theta\, e^{\pm i\phi}$
	±2	$\left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\, e^{\pm 2i\phi}$

Orbital with **orbital angular momentum quantum number** ***l*** has an angular momentum of magnitude $\{l(l+1)\}^{1/2}\hbar$, with $l = 0, 1, 2, \ldots, n-1$.

Orbital with **magnetic quantum number** ***m*** has a *z*-component of angular momentum $m\hbar$, with $m = 0, \pm 1, \pm 2, \ldots, \pm l$.

Hydrogenic Atoms

Radial wave equation

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2}+V_{\text{eff}}u=Eu \qquad u=rR$$

$$E_n=-\frac{Z^2\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2n^2}$$

Principal quantum number ***n***, determines the energy, ***n* = 1, 2, 3, ...**

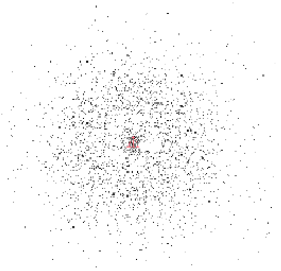
Orbital	n	l	$R_{n,l}$
1*s*	1	0	$2\left(\frac{Z}{a}\right)^{3/2}e^{-\rho/2}$
2*s*	2	0	$\frac{1}{8^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(2-\rho)e^{-\rho/2}$
2*p*	2	1	$\frac{1}{24^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho e^{-\rho/2}$
3*s*	3	0	$\frac{1}{243^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(6-6\rho+\rho^2)e^{-\rho/2}$
3*p*	3	1	$\frac{1}{486^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(4-\rho)\rho e^{-\rho/2}$
3*d*	3	2	$\frac{1}{2430^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho^2e^{-\rho/2}$

$$\rho=\frac{2Zr}{na}$$

Bohr radius

$$a=\frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}=0.529 \quad \text{Å}$$

Atomic orbital and energy



An atomic orbital: a one-electron wavefunction for an electron in an atom, defined by **the 3 quantum numbers, *n*, *l*, *m*.**

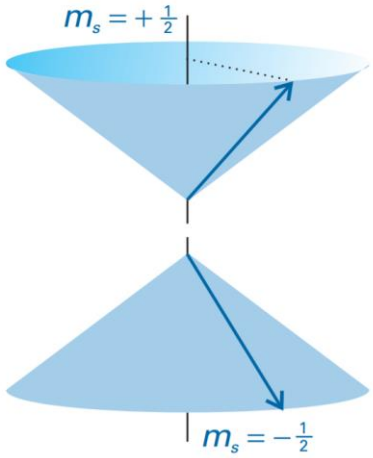
Principal quantum number（主量子数） ***n*,** → the energy of the orbital, ***n* = 1, 2, 3, ...**

Angular momentum quantum numbers (角量子数) *l*, *m* specify the angular momentum of the electron around the nucleus.

Angular momentum of magnitude $\{l(l+1)\}^{1/2}\hbar$, with $l = 0,1, 2, \ldots, n-1$.

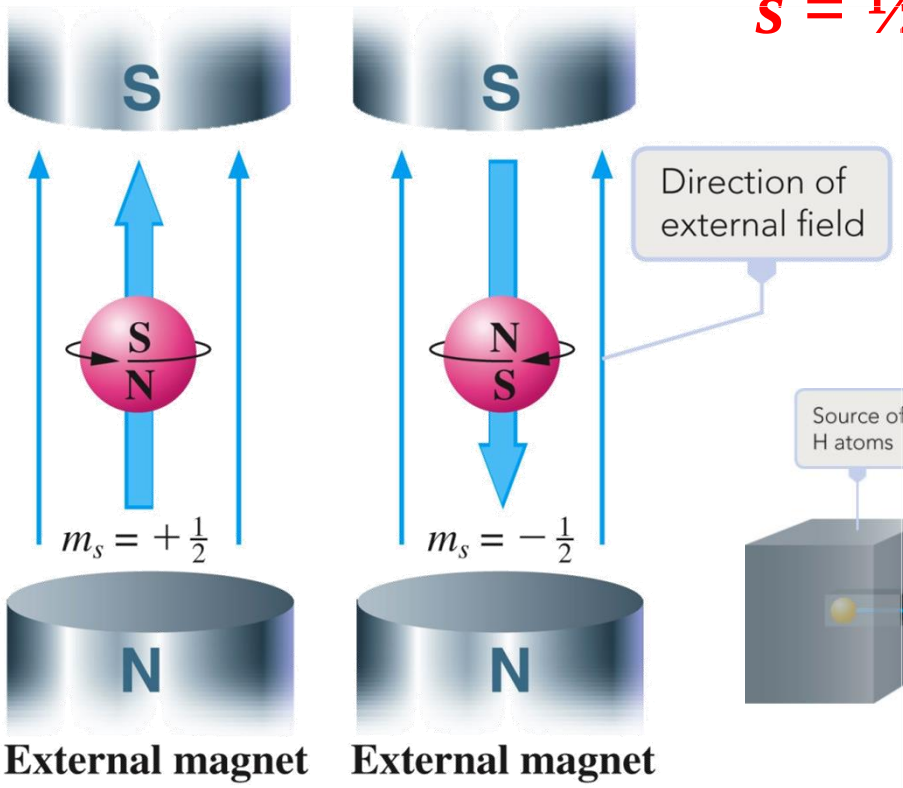
z-component of angular momentum $m\hbar$, with $m = 0, \pm1, \pm 2, \ldots, \pm l$.

Specify a state of an electron, n, l, m_l, m_s

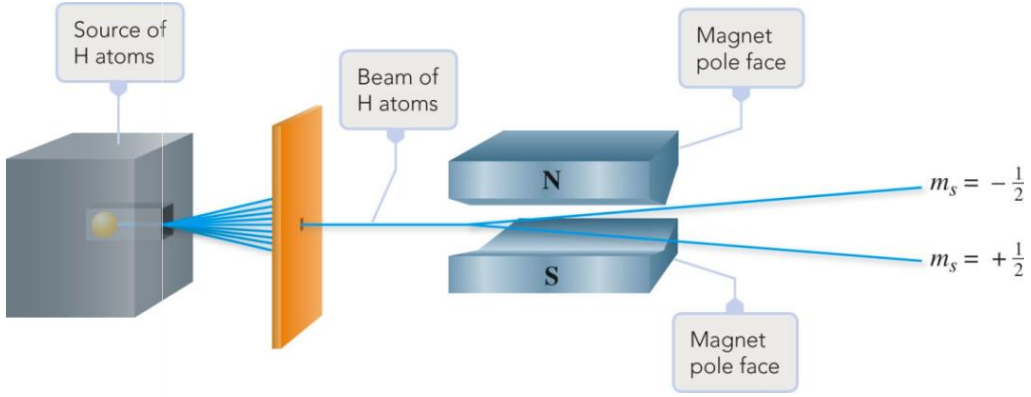


Intrinsic spin angular momentum quantum numbers (自旋角量子数) of electron, s, m_s

$s = ½$, $m_s = \pm ½$



An electron in an atom behaves like a tiny magnet and orients itself in a magnetic field.



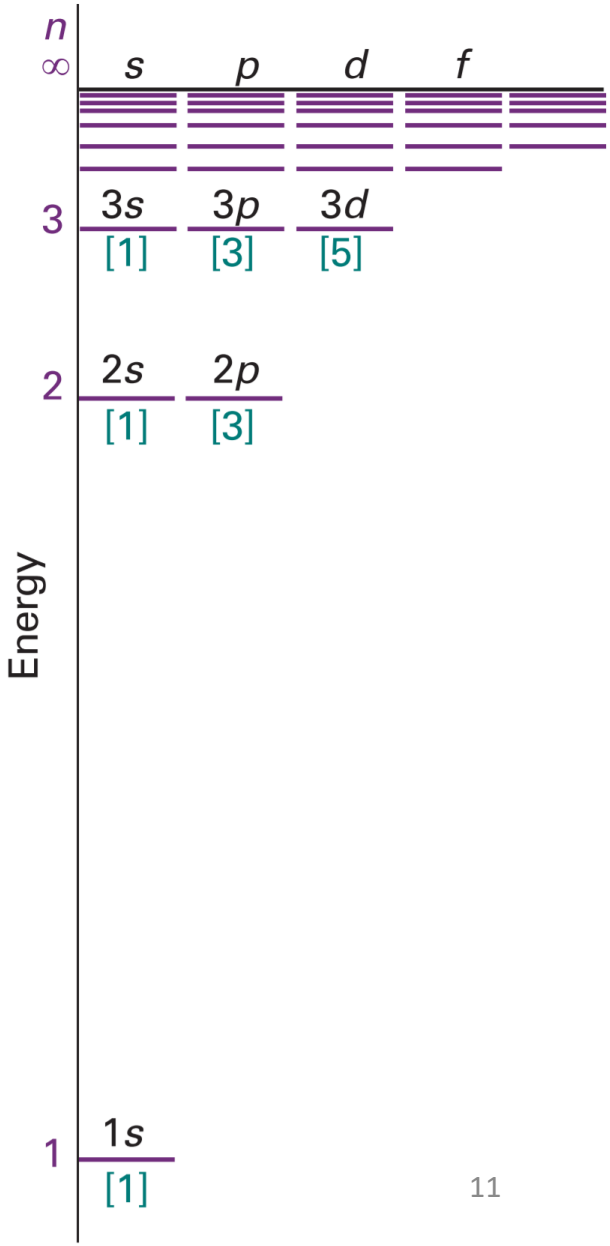
The energy levels

$$E_n = -\left(\frac{Z^2 \mu e^4}{32\pi^2 \varepsilon_0^2 \hbar^2}\right)\frac{1}{n^2} \qquad n = 1, 2, \ldots$$

The higher the value of "*n*" , the greater the energy of the orbital.

The **degeneracy** of the energy level characterized by *n* is $\boldsymbol{n^2}$

The energy levels of a H atom



Shells and Subshells

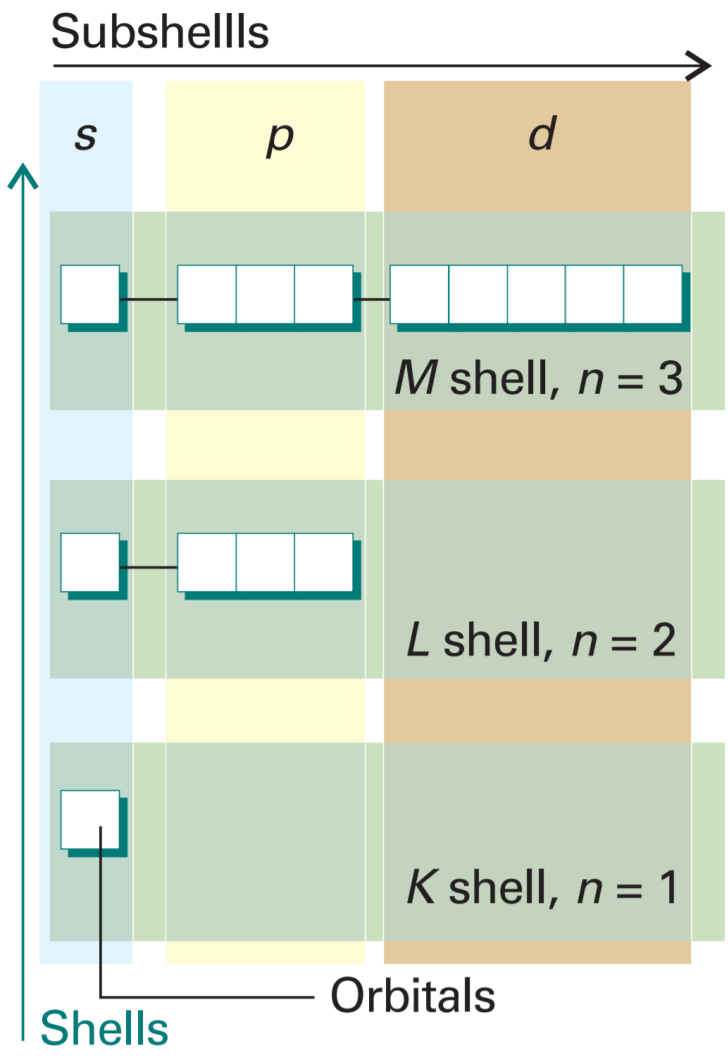
Shell: all the orbitals with the same *n*.

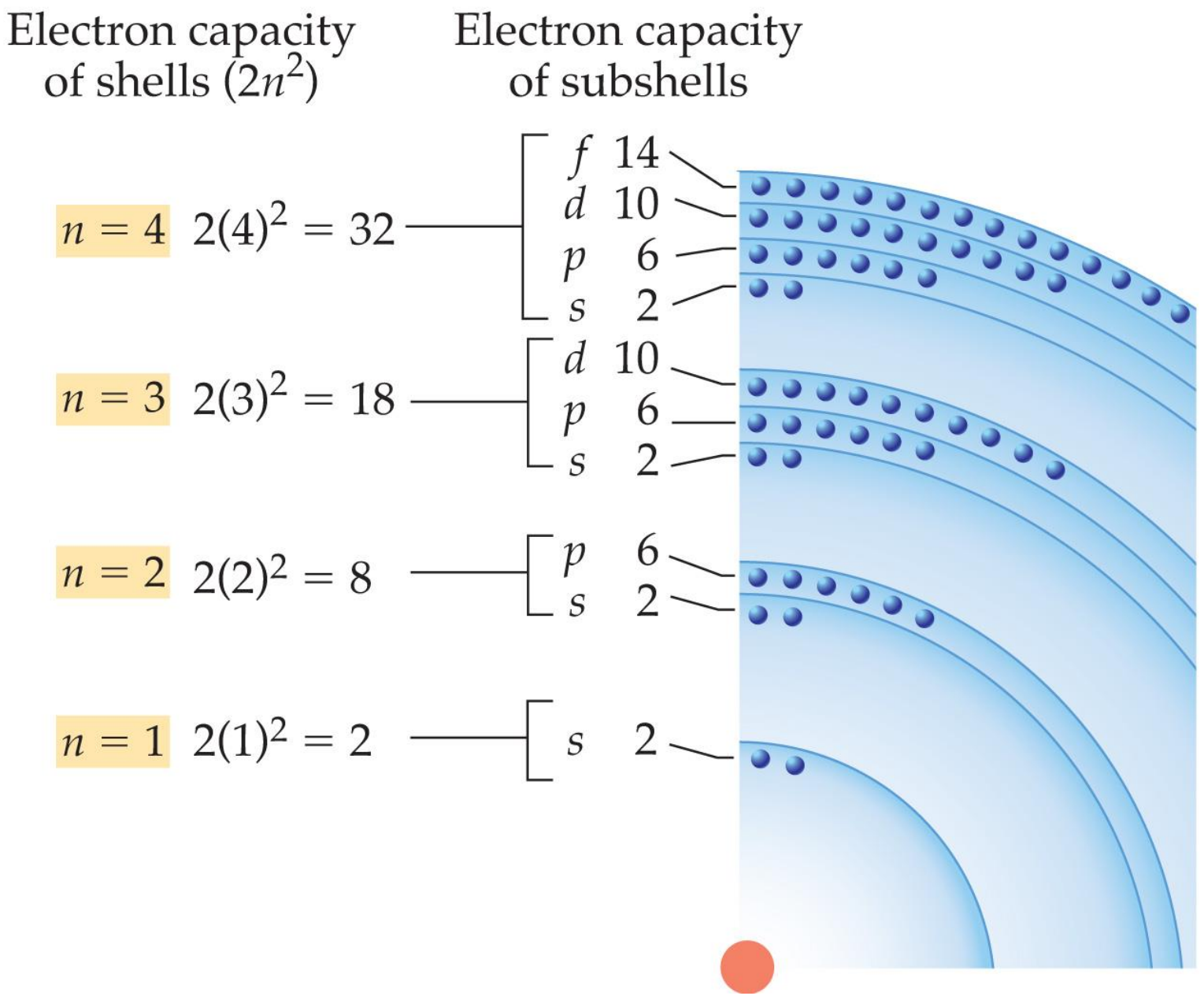
$n =$	1	2	3	4 . . .
	K	L	M	N . . .

Subshell: all the orbitals with the same *n* but different of *l* form a subshell of the given shell.

$l =$	0	1	2	3
	s	p	d	f

4	5	6 . . .
g	h	i . . .



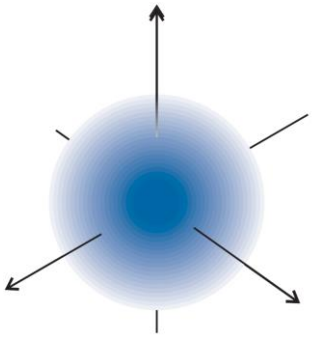

Electron capacity of shells ($2n^2$)
Electron capacity of subshells
$n = 4$ $2(4)^2 = 32$
f 14
d 10
p 6
s 2
$n = 3$ $2(3)^2 = 18$
d 10
p 6
s 2
$n = 2$ $2(2)^2 = 8$
p 6
s 2
$n = 1$ $2(1)^2 = 2$
s 2

s atomic orbital

l,	m,	$Y_{l,m}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$

Spherically symmetrical

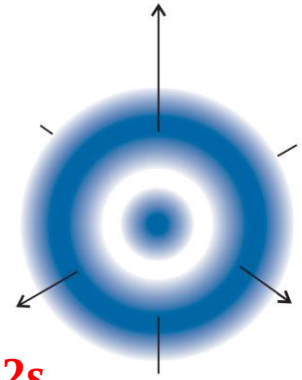
Density of shading



1*s*

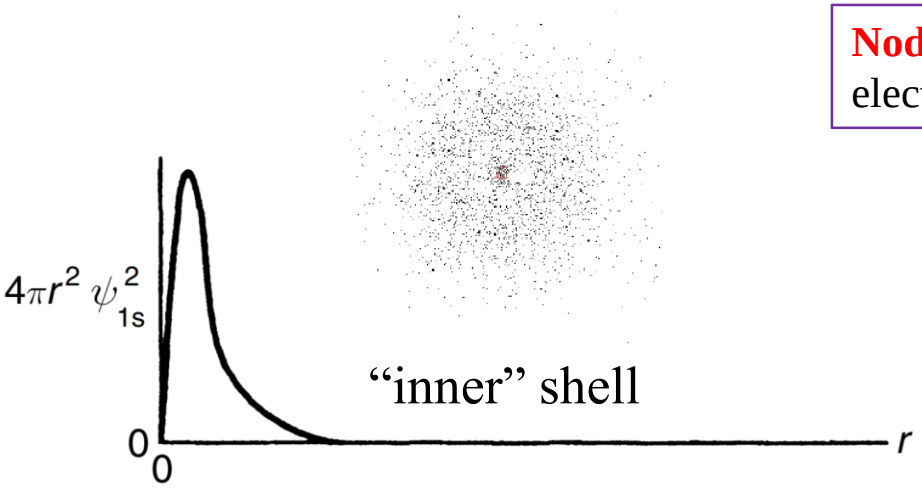
The orbital with $n=1$: $l=0$, $m_l=0$ **1*s***

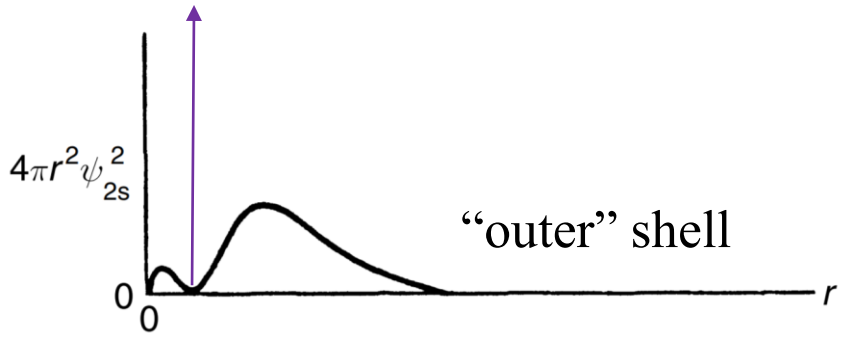
$n=2$: $l=0$, $m_l=0$ **2*s***



2*s*

Node: a surface on which an electron is not found.

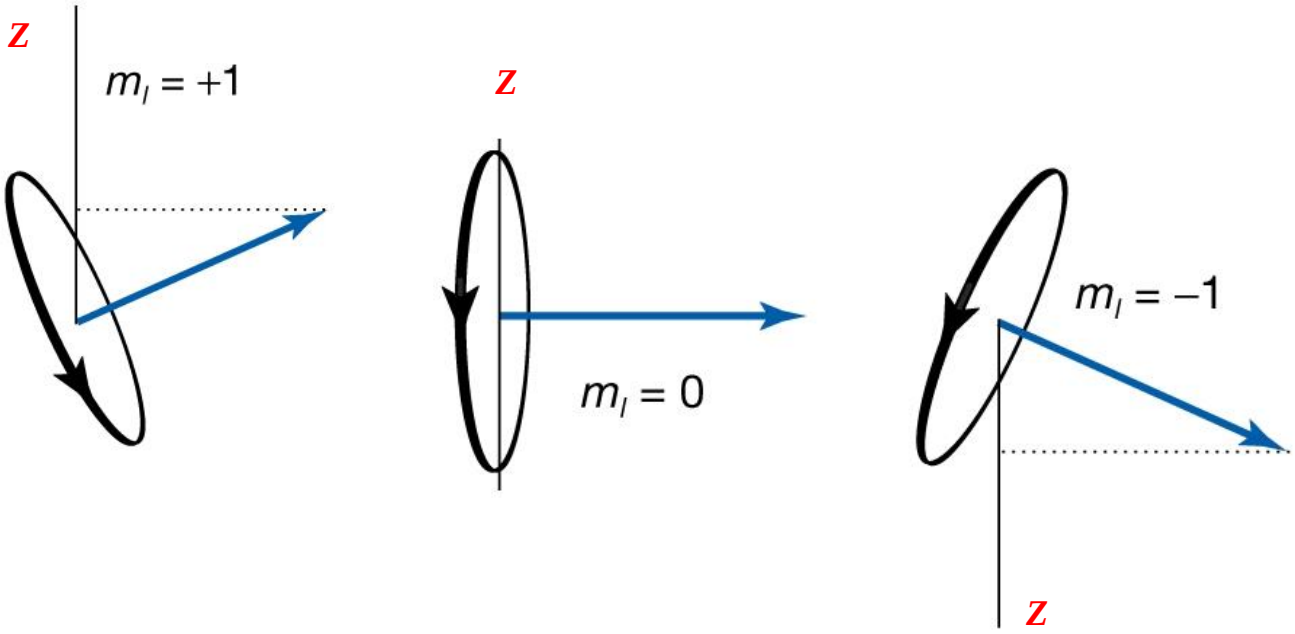


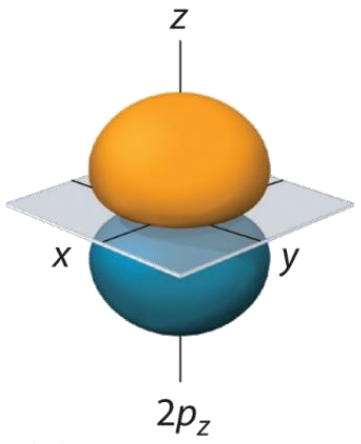


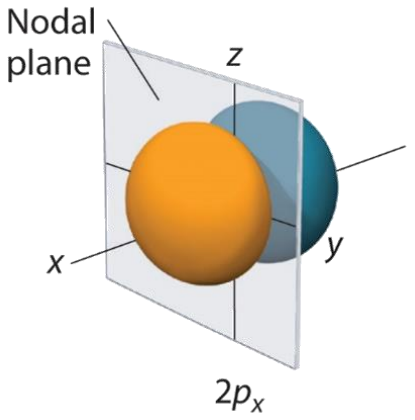
p atomic orbital $l = 1$

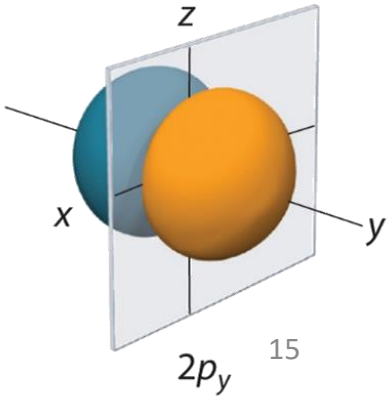
The orbital with $n=2$: $l=1$, $m_l=0$ **$2p_0$ or $2p_z$**

$n=2$: $l=1$, $m_l=\pm1$ **$2p_{\pm1}$ or $2p_x$, $2p_y$**

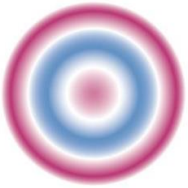




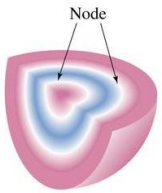


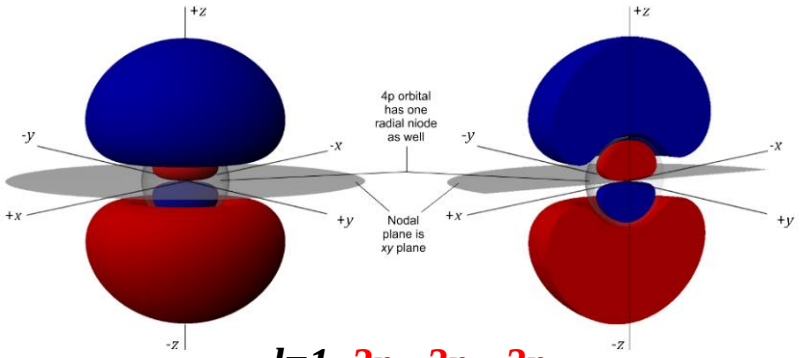


Atomic orbitals with *n*=3



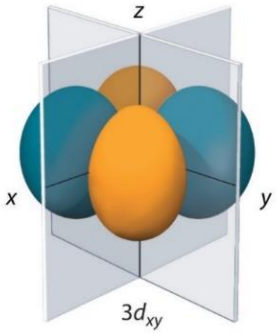
l=0, 3s

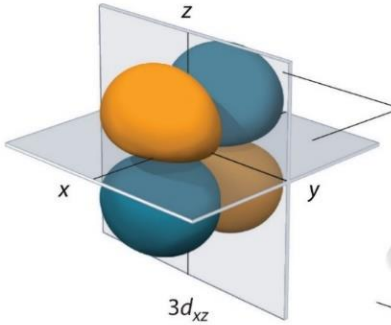


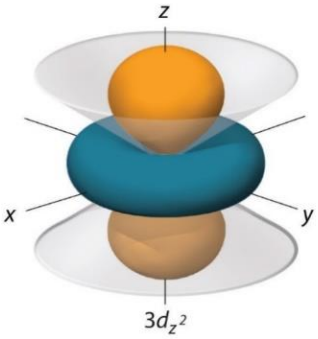


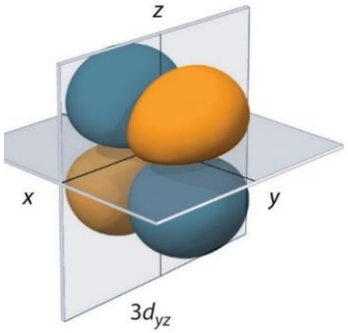
l=1, $3p_z$, $3p_x$, $3p_y$

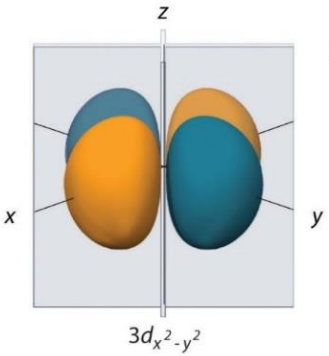
l=2, m=-2,-1,0,1,2,
$3d_{z^2}$, $3d_{xy}$, $3d_{xz}$, $3d_{yz}$, $3d_{x^2-y^2}$











Chapter 1

Quantum Theory

Part IV

Contents

- Structures of Many-Electron Atoms
- Molecular Structure

Structures of many-electron atoms

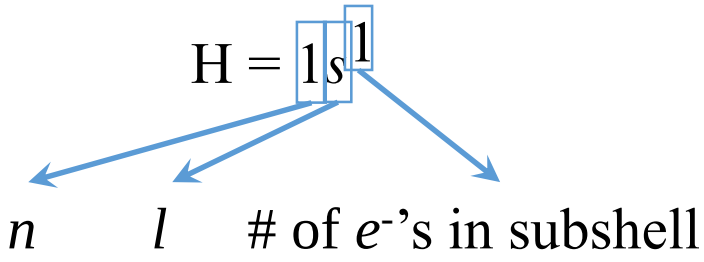
Orbital Approximation: think of each electron as occupying its 'own' orbital.

$$\psi(r_1, r_2, \ldots) = \psi(r_1)\psi(r_2)\ldots$$

Hydrogenic orbital corresponding to **effective nucluear charges Z_{eff}**

Electronic Configuration (电子组态): the list of occupied orbitals.

The ground state of a hydrogenic atom with a single electron in a 1s orbital: **$1s^1$**.



More than one electron , i.e. He atom ?

$$\psi(r_1, r_2, \ldots) = \psi(r_1)\psi(r_2)\ldots$$

He atom ?

Pauli exclusion principle (泡利不相容原理):
No more than two electrons may occupy any given orbital, and if two do occupy one orbital, then their spins must be paired.

The ground state of He atom: **$1s^2$**.

He = $\frac{1\uparrow}{1s}$ or $\frac{1\downarrow}{1s}$





↑, α spin **↓, β spin**

"orbital filling diagram"

The exclusion principle is the key to the structure of complex atoms, to chemical periodicity, and to molecular structure.

Building-up principle

For a ground state:

a. **Pauli exclusion principle**: Every orbital may contain only two electrons of opposite spins.

b. **The principle of minimum energy**: Whilst being compatible with the Pauli principle, electrons occupy the orbital with the lowest energy first.

c. **Hund's rule**: In degenerate energy states, electrons tend to occupy as many degenerate orbitals as possible. (Adopt a configuration with the greatest number of unpaired electrons.)

Electrons occupy the orbitals in the order of:

$1s$ $2s$ $2p$ $3s$ $3p$ $4s$ $3d$ $4p$ $5s$ $4d$ $5p$ $6s$

Penetration (钻穿) and shielding (屏蔽)

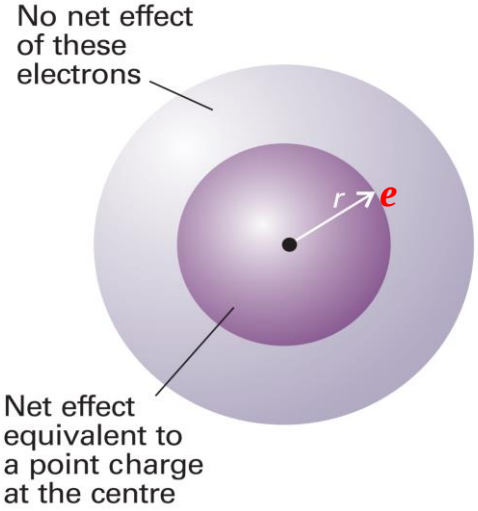
All subshells of a given shell in hydrogenic atoms: are degenerate

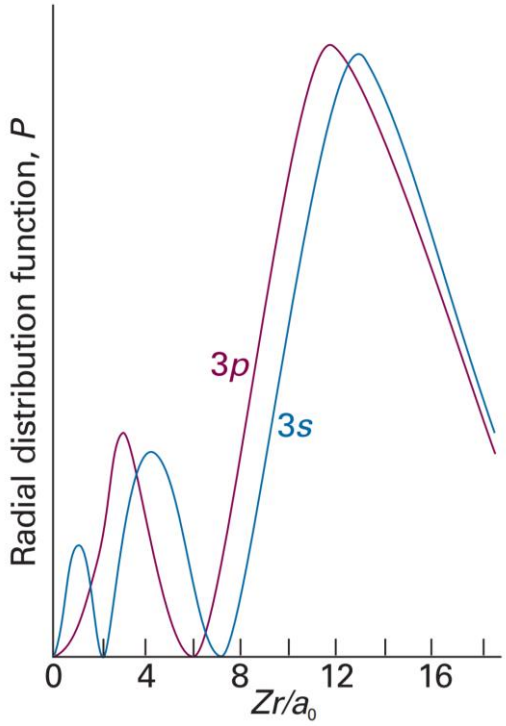
in many-*e* atoms: are not degenerate

because of **shielding** of outer electrons

Effective nuclear charge $Z_{eff} = Z - \sigma$

Shielding constant





Penetration

$s > p > d > f$ electron

Energy order of subshells of a given shell: $s < p < d < f$

Electrons occupy the orbitals in the order of:

$1s$ $2s$ $2p$ $3s$ $3p$ $4s$ $3d$ $4p$ $5s$ $4d$ $5p$ $6s$

The ground-state configuration of C: $1s^2 2s^2 2p^2$ = [He] $2s^2 2p^2$ → $1s^2 2s^2 2p^1 2p^1$

N: [He] $2s^2 2p_x^1 2p_y^1 2p_z^1$

O: [He] $2s^2 2p_x^2 2p_y^1 2p_z^1$

The ground-state configuration of Ne: [He]$2s^2 2p^6$ → [Ne]

Na: [Ne] $3s^1$

Ar: [Ne] $3s^2 3p^6$ → [Ar]

Sc: [Ar] $3d^1 4s^2$ ⇨ Strong *e*-*e* repulsions in the 3*d* orbitals are minimized.

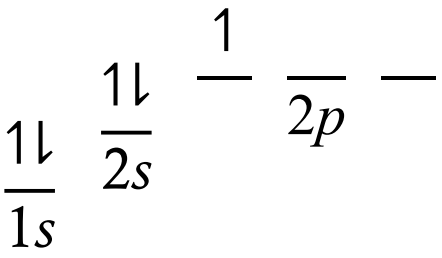
Sc-Zn (except for Cr and Cu): [Ar] $3d^n 4s^2$

Cr: [Ar] $3d^5 4s^1$

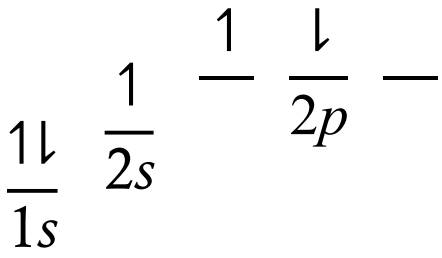
Cu: [Ar] $3d^{10} 4s^1$

Ground & Excited States

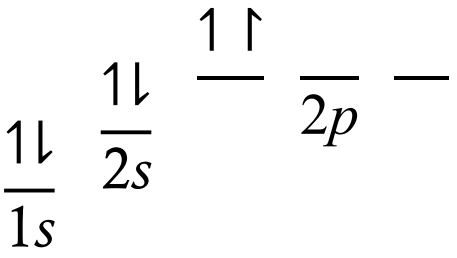
- The electron configurations we've discussed so far have all been *ground states*.
  - Ground State – Lowest energy electron configuration for an element.
- Any other valid configuration of electrons for a given element represents an *excited state*.
  - An element will only have one ground state, but it can have many excited states!



Ground State



Excited State



Invalid Configuration

Atomic terms (原子谱项)

A given electron configuration gives rise in general to several different atomic states.

The total spin angular momentum quantum number ***S***

$^{2S+1}L_J$

Multiplicity

The total orbital angular momentum quantum number ***L***

The total angular momentum quantum number ***J***

$(2L + 1)(2S + 1)$ terms

***L* ⇒ *2L+1* orientations, M_L=*L, L-1, …, -L, L* is a non-negative integer**

$L = l_1 + l_2, l_1 + l_2 - 1, \ldots, |l_1 - l_2|$ p^2 configuration ⇒ l_1=l_2=1, so *L*=2,1,0 ⇒ D, P, S

L:	0	1	2	3	4	5	6 . . .
	S	P	D	F	G	H	I . . .

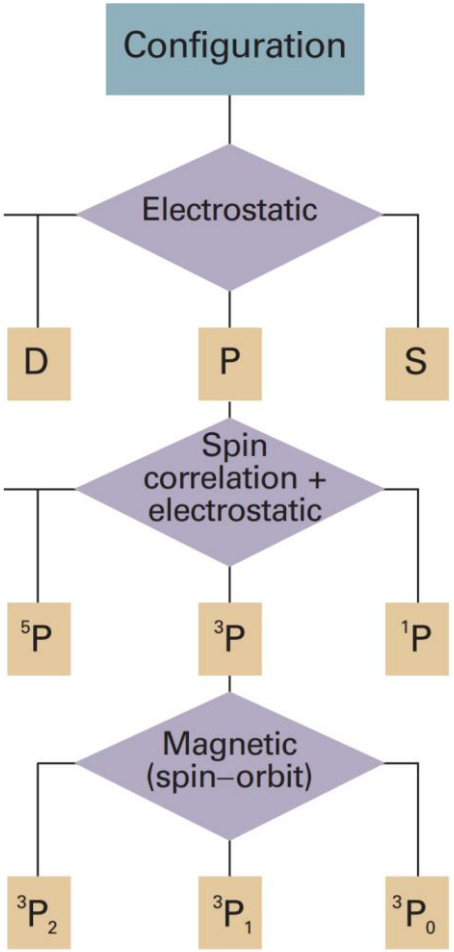
***S* ⇒ each *e* has *s*=1/2, *S* is a non-negative integer or half integer**

$S = s_1 + s_2, s_1 + s_2 - 1, \ldots, |s_1 - s_2|$ p^2 configuration ⇒ s_1=s_2=1/2, so *S*=1,0 ⇒ 2*S*+1=3,1

For light atoms $J = L + S, L + S - 1, \ldots, |L - S|$

^{2}P term ⇒ $^2P_{1/2}, ^2P_{3/2}$; ^{3}D term ⇒ $^3D_3, ^3D_2, ^3D_1$; ^{2}S term ⇒ $^2S_{1/2}$

Convention: lowercase letters to label orbitals; uppercase letters to label overall states.



$$L = l_1 + l_2, l_1 + l_2 - 1, \ldots, |l_1 - l_2|$$

$$S = s_1 + s_2, s_1 + s_2 - 1, \ldots, |s_1 - s_2|$$

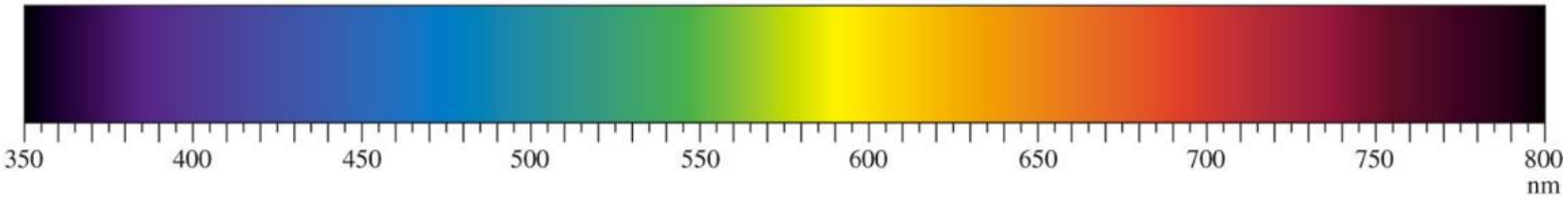
$$J = L + S, L + S - 1, \ldots, |L - S|$$

$$^{2S+1}L_J$$

Atomic Spectra

A **transition** of electron ⇒ a change of state, a change of energy

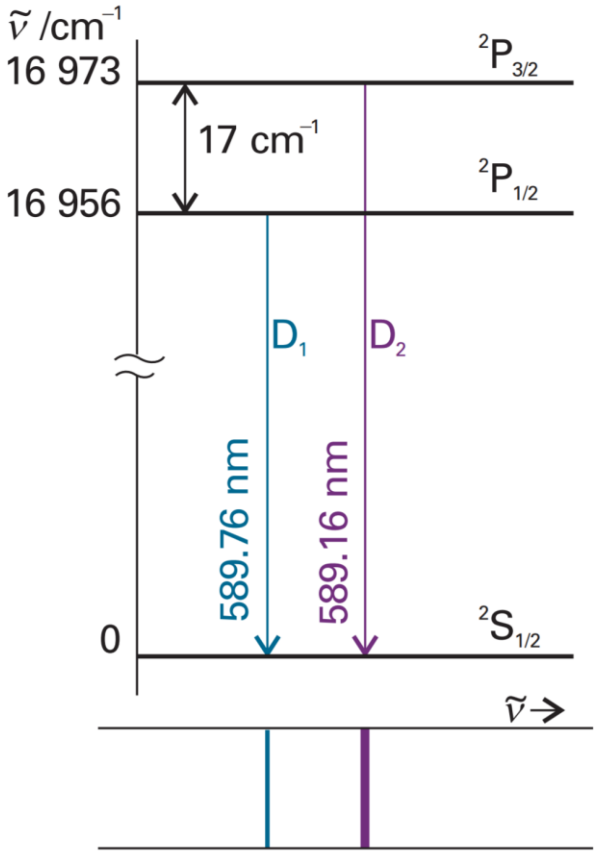
- Electrons in an atom can absorb energy, moving from a lower to a higher energy level. Referred to as an *excited state*.
- The electron can then release the energy to go back to its original, lowest energy state. Referred to as the *ground state*.
- The light emitted can be dispersed through a prism or diffraction grating into its component wavelengths (colors).



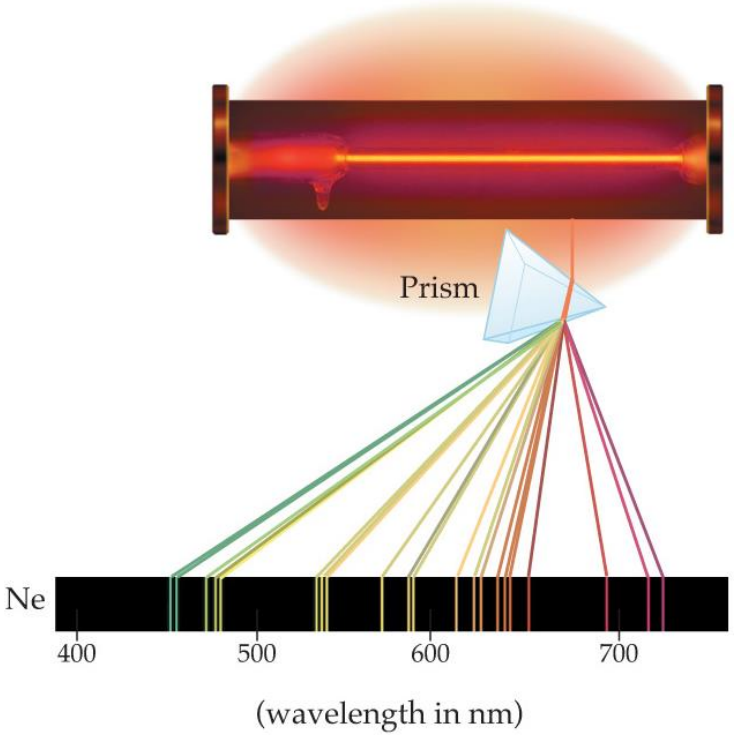
Ordinary sunlight or incandescent light gives a continuous spectrum of light.

Atomic Spectra

Emission spectrum



- Excited atoms in a vacuum only emit certain, characteristic wavelengths of light.
  - Called *line spectra*.
- The pattern of emission is different for every element.
  - These spectra serve as "fingerprints" for identification of elements.



Spectroscopic transitions and selection rules

Selection rules for spectral transition *in light atoms*

$\Delta S=0$; $\Delta L=0,\pm1$; $\Delta l=\pm1$;

$\Delta J=0,\pm1$, but $J=0$ to $J=0$ is forbidden

Based on the following facts:

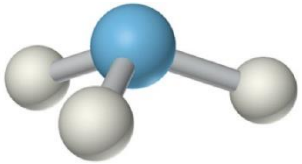
- conservation of angular momentum;
- a photon has a spin of 1;
- the light does not affect the spin directly;
- the orbital angular momentum of an individual electron must change

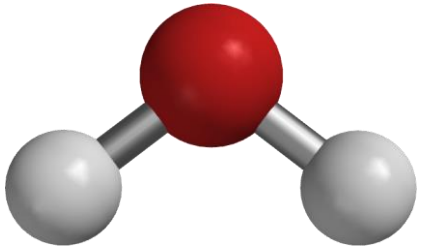
In heavy atoms when *jj*-coupling is appropriate,

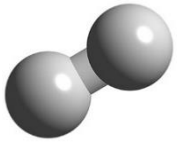
$\Delta S=\pm1$ are allowed.

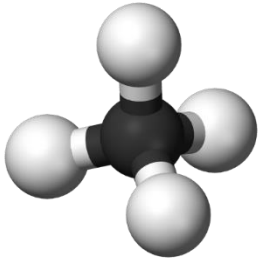
Molecular Structure

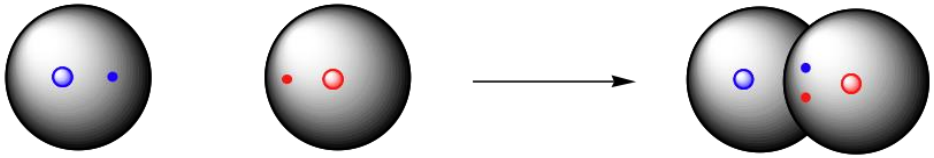
- The Born–Oppenheimer approximation
- Valence-bond theory
- Molecular orbital theory

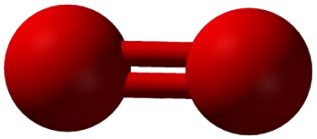






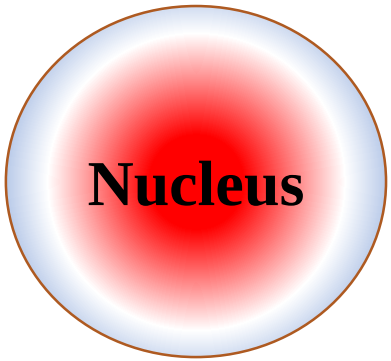






The Born-Oppenheimer Approximation

is supposed that the nuclei, being so much heavier than an electron, move relatively slowly and may be treated as stationary while the electrons move in their field.



Moves around very fast



Electron





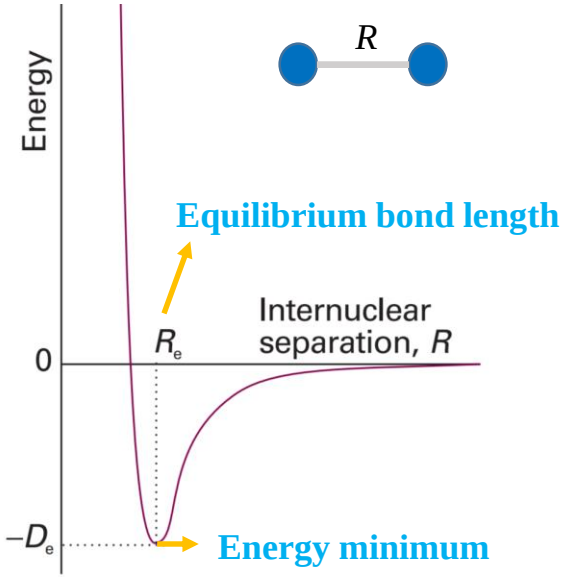


Too heavy, relate to electron, just like at rest.



Solve the Schrödinger equation of electrons alone

The Potential Energy Surface



Select an *R* and then to solve the Schrödinger equation

